

# Brazilian E-commerce Sales Analysis

## Final Business Insight Report

### Project Overview

This project analyzes the **Brazilian E-commerce Public Dataset by Olist** to understand marketplace sales performance, customer purchasing behavior, product performance, payment preferences, seller contribution, geographic demand, and delivery performance.

The analysis follows an end-to-end data analytics workflow using **Python, SQL, and Power BI**.

The primary objective is to transform transactional e-commerce data into actionable business insights that can support decisions related to sales growth, customer experience, product strategy, seller management, and logistics.

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## 1. Executive Summary

The analysis provides a comprehensive view of Olist's e-commerce marketplace performance across sales, customers, products, payments, sellers, geography, and delivery.

The findings indicate that e-commerce performance should not be evaluated using sales volume alone. A complete assessment requires understanding **where sales are generated, which products and sellers drive revenue, how customers purchase, how they pay, and how efficiently orders are delivered**.

The analysis also highlights the importance of monitoring delivery performance because the customer experience extends beyond the purchase itself. Strong sales combined with inefficient delivery can create operational and customer-experience risks.

Overall, the analysis provides a data-driven framework for identifying high-performing market segments, improving operational efficiency, and supporting future marketplace growth.

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## 2. Data & Analysis Scope

The analysis combines information from multiple Olist datasets covering:

- Orders
- Customers

- Sellers
- Products
- Product categories
- Payments
- Reviews
- Order timestamps
- Delivery information
- Customer and seller locations

The original dataset contains approximately 100K orders and multiple related transactional tables.

For the main delivery and sales analysis, the project focuses on **delivered orders**, because delivery-related metrics should primarily be calculated from completed customer deliveries.

The original order-status distribution showed:

| Order Status | Orders  |
|--------------|---------|
| Delivered    | 113,384 |
| Shipped      | 1,214   |
| Canceled     | 549     |
| Processing   | 362     |
| Invoiced     | 359     |
| Unavailable  | 7       |
| Approved     | 3       |

The analysis therefore uses delivered orders as the primary population for completed-order performance analysis.

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## 3. Business Question Framework

The analysis was designed around the following business questions:

### Sales

- How is sales performance changing over time?
- Which periods generate the highest sales?
- What is the average order value?

### Customers

- How do customers behave across the marketplace?
- Which locations generate the greatest demand?
- What patterns can be observed in customer purchasing?

## **Products**

- Which product categories generate the greatest sales?
- Which categories have the strongest demand?
- Where are opportunities for product-level growth?

## **Payments**

- Which payment methods are most commonly used?
- How does payment behavior affect the customer purchasing experience?

## **Sellers**

- Which sellers contribute most to marketplace sales?
- Is marketplace performance concentrated among a relatively small group of sellers?

## **Logistics**

- How long does it take for customers to receive their orders?
- How is delivery performance distributed?
- Where might logistics improvements create a better customer experience?

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# **4. Key Business Insights**

## **Insight 1 — Sales Performance Should Be Monitored Over Time**

Monthly sales analysis provides an important view of marketplace growth and demand patterns.

Rather than evaluating the business from total sales alone, management should monitor sales trends month by month to identify:

- Growth periods
- Declining periods
- Seasonal demand
- Changes in purchasing activity
- Potential unusual fluctuations

### **Business implication**

A monthly sales trend can be used as an early-warning indicator.

If sales begin to decline for several consecutive periods, management can investigate whether the cause is related to:

- Product availability
- Customer demand
- Pricing
- Seller activity
- Marketing
- Payment behavior
- Delivery performance

### **Recommendation**

Create a regular sales-monitoring process using monthly revenue, order volume, and AOV together rather than relying on revenue alone.

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## **5. Insight 2 — Product Category Performance Is Uneven**

The product-category analysis demonstrates that customer demand is not distributed equally across all categories.

Some categories contribute substantially more to marketplace sales and order volume than others.

### **Business implication**

High-performing categories represent important opportunities for:

- Inventory expansion
- Seller acquisition
- Promotional campaigns
- Cross-selling
- Category-specific marketing

At the same time, low-performing categories should not automatically be discontinued. Their performance should be evaluated alongside:

- Average order value
- Order frequency
- Customer demand
- Product availability
- Delivery performance
- Profitability, if cost data is available

## **Recommendation**

Segment product categories into:

### **High-performing**

→ Maintain availability and explore expansion.

### **Growth opportunity**

→ Investigate customer demand and promotional potential.

### **Low-performing**

→ Review pricing, assortment, availability, and customer demand before making strategic decisions.

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## **6. Insight 3 — Geographic Demand Creates Opportunities for Regional Strategy**

Sales-by-state analysis shows that customer demand varies geographically across Brazil.

This means that the marketplace should not treat the entire country as a single homogeneous market.

### **Business implication**

High-demand states can receive greater focus through:

- Seller recruitment
- Product availability
- Regional marketing
- Faster fulfillment
- Customer acquisition campaigns

Lower-demand regions may require additional investigation before significant investment.

## **Recommendation**

Develop regional performance monitoring using:

- Sales
- Orders
- Customers
- AOV
- Delivery time
- Seller availability

This would help identify regions with strong demand but insufficient seller or logistics coverage.

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## **7. Insight 4 — Payment Preferences Are Important for Conversion**

The payment analysis shows that customers use multiple payment methods, with digital payment behavior playing an important role in marketplace transactions.

Payment methods analyzed include:

- Credit card
- Boleto
- Voucher
- Debit card

### **Business implication**

A marketplace should make popular payment methods easy and reliable for customers.

Payment friction can negatively affect the purchasing experience.

### **Recommendation**

Management should:

- Maintain reliable support for the most-used payment methods.
  - Monitor payment failures.
  - Analyze payment behavior by customer segment.
  - Examine installment behavior.
  - Evaluate whether different payment methods are associated with different order values.
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## 8. Insight 5 — Seller Performance Is Concentrated

Seller-level analysis allows the marketplace to identify sellers that contribute significantly to sales and order volume.

This is important because marketplace performance depends not only on customer demand but also on seller participation and performance.

### **Business implication**

High-performing sellers can be strategically important partners.

The marketplace can potentially support them through:

- Seller retention programs
- Performance incentives
- Promotional opportunities
- Better operational support
- Seller-specific analytics

At the same time, sellers with weak performance should be monitored for:

- Delivery problems
- Low order volume
- Product availability
- Customer dissatisfaction
- Operational issues

### **Recommendation**

Create a seller-performance scorecard containing:

- Sales
- Orders
- AOV
- Delivery performance
- Review performance
- Cancellation rate

This would provide a more complete view than ranking sellers by sales alone.

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## 9. Insight 6 — Delivery Performance Is a Critical Customer Experience Metric

Delivery analysis is one of the most important operational components of this project.

The analysis calculates delivery time using order and delivery timestamps.

### **Business implication**

The purchase does not end when a customer places an order.

The customer experience continues through:

**Purchase → Processing → Shipping → Delivery**

Long delivery times can potentially affect customer satisfaction and future purchasing behavior.

### **Recommendation**

Management should monitor:

- Average delivery time
- Median delivery time
- Delivery-time distribution
- Extremely delayed orders
- Delivery performance by state
- Delivery performance by seller
- Delivery performance by product category

The median should be considered alongside the average because a small number of extremely delayed orders can influence the mean.

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## 10. Insight 7 — Freight Cost Should Be Considered Alongside Product Sales

The analysis includes both product price and freight value.

A more complete view of transaction value is therefore:

**Total Order Value = Product Price + Freight Value**



## Business implication

A product that generates high sales revenue may still have a different customer value proposition depending on its freight cost.

High freight charges can become particularly important for:

- Large products
- Heavy products
- Long-distance deliveries
- Low-value products

## Recommendation

Analyze freight value by:

- Product category
- Customer state
- Seller state
- Product weight
- Order value

This can identify situations where logistics costs may be affecting purchasing behavior.

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# 11. Insight 8 — Average Order Value Provides a Better View of Customer Value

Total sales and order volume should be interpreted together with **Average Order Value (AOV)**.

## Formula

**AOV = Total Order Value ÷ Number of Orders**

A rising order count does not necessarily mean customers are spending more per transaction.

Likewise, increasing AOV with declining order volume may indicate that fewer customers are making larger purchases.

## Recommendation

Track three metrics together:

**Revenue + Orders + AOV**

This provides a much stronger view of marketplace performance than any single metric.

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## **12. Insight 9 — Customer Purchase Behavior Should Be Examined Beyond Total Customers**

The customer analysis should distinguish between the number of customers and how frequently those customers purchase.

A marketplace can grow through:

- Acquiring new customers
- Increasing repeat purchases
- Increasing order value
- Improving customer retention

### **Business implication**

Customer acquisition alone does not guarantee sustainable growth.

Repeat purchasing can indicate stronger customer engagement and marketplace loyalty.

### **Recommendation**

Monitor:

- New customers
- Repeat customers
- Orders per customer
- AOV by customer segment
- Repeat purchase rate
- Customer geographic distribution

This can support more targeted customer-retention strategies.

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## 13. Insight 10 — Dashboard-Based Monitoring Can Improve Decision Making

The Power BI dashboard combines multiple dimensions of the analysis into an interactive business intelligence environment.

Users can examine performance through:

- Time
- Geography
- Product category
- Payment method
- Seller
- Customer behavior
- Delivery performance

### **Business implication**

Instead of manually analyzing large transactional datasets, decision-makers can use the dashboard to quickly identify changes and investigate specific segments.

### **Recommendation**

The dashboard should be used as a management-monitoring tool rather than simply as a visualization.

Regular KPI monitoring can help identify:

- Sales changes
  - Category performance changes
  - Regional opportunities
  - Delivery problems
  - Seller performance issues
  - Customer behavior changes
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## 14. Strategic Recommendations

Based on the overall analysis, the following strategies are recommended.

### **A. Strengthen High-Performing Categories**

Identify categories that consistently generate strong sales and order volume.

Actions:

- Maintain product availability
  - Recruit more sellers in strong categories
  - Use targeted promotions
  - Analyze cross-selling opportunities
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## **B. Improve Delivery Performance**

Delivery is a major component of customer experience.

Actions:

- Monitor delivery time by state
  - Identify slow-performing sellers
  - Investigate logistics bottlenecks
  - Monitor extreme delivery delays
  - Compare expected and actual delivery performance when estimated delivery data is available
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## **C. Develop Regional Strategies**

Use state-level performance to identify:

- High-demand markets
- Underserved markets
- Logistics opportunities
- Seller recruitment opportunities

Regional strategies should combine sales and delivery metrics.

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## **D. Strengthen Seller Management**

Create seller-performance monitoring based on multiple KPIs.

Recommended seller KPIs:

- Sales
- Orders
- AOV
- Delivery time
- Review score
- Cancellation rate

High-performing sellers can be retained and supported, while underperforming sellers can receive targeted operational attention.

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## **E. Improve Customer Retention**

Customer analytics should focus not only on acquisition but also on repeat purchasing.

Actions:

- Identify repeat customers
  - Analyze purchasing frequency
  - Identify high-value customers
  - Develop personalized promotional strategies
  - Investigate factors associated with repeat purchasing
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## **F. Optimize Payment Experience**

Continue supporting popular payment methods while monitoring payment behavior by:

- Customer segment
- Order value
- Product category
- Geography

This can help reduce friction throughout the purchasing process.

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# **15. Recommended KPI Framework**

For ongoing business monitoring, the following KPI structure is recommended.

### **Sales KPIs**

- Total Sales
- Total Orders
- Average Order Value
- Monthly Sales Growth
- Sales by Category

### **Customer KPIs**

- Total Customers
- Repeat Customers
- Orders per Customer
- Repeat Purchase Rate

### **Product KPIs**

- Top Categories
- Category Sales
- Category Order Volume

### **Seller KPIs**

- Total Sellers
- Seller Sales
- Seller Orders
- Seller AOV
- Seller Delivery Performance

### **Logistics KPIs**

- Average Delivery Days
- Median Delivery Days
- Delayed Orders
- Delivery Performance by Region

### **Payment KPIs**

- Payment Method Share
- Average Order Value by Payment Method
- Installment Distribution

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## **16. Limitations of the Analysis**

Several limitations should be considered when interpreting the results.

## **1. Historical Dataset**

The dataset represents historical marketplace transactions rather than current Olist operations.

Therefore, the findings should be interpreted as analytical and educational rather than as a current business performance report.

## **2. Revenue vs. Profit**

The analysis focuses primarily on sales/order value.

Because complete business-cost information is not available, the project should not interpret high sales as high profitability.

## **3. Delivered-Order Focus**

Delivery analysis primarily uses completed/delivered orders.

This is appropriate for delivery-time calculations but means that incomplete, canceled, or other order statuses require separate analysis.

## **4. Customer Identification**

Customer behavior analysis depends on the customer identifiers and order structure available in the dataset.

Further customer-lifetime analysis would require additional assumptions and modeling.

## **5. Missing Values**

Several fields contained missing values during the cleaning process, particularly delivery-related fields and a small number of product attributes.

These were investigated and handled as part of the preprocessing stage.

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# **17. Final Business Conclusion**

The Brazilian Olist e-commerce dataset demonstrates how transactional data can be transformed into meaningful business intelligence.

The analysis shows that a marketplace should evaluate performance from several perspectives simultaneously:

**Sales → Customers → Products → Sellers → Payments → Geography → Delivery**

The most important business lesson is that **sales growth alone does not provide a complete picture of marketplace health.**

A sustainable e-commerce strategy should combine:

- Strong product and category performance
- Healthy customer purchasing behavior
- Effective seller participation
- Convenient payment options
- Efficient delivery
- Strong regional coverage
- Continuous KPI monitoring

The Power BI dashboard provides an interactive way to monitor these dimensions, while Python and SQL provide the analytical foundation behind the dashboard.

This project demonstrates an end-to-end data analytics workflow:

**Raw Data → Data Cleaning → EDA → SQL Analysis → KPI Development → Power BI Dashboard → Business Insights → Recommendations**

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## 18. Portfolio Value

This project demonstrates practical experience in:

- Data cleaning and preprocessing
- Exploratory Data Analysis
- Business question formulation
- SQL-based analysis
- KPI development
- DAX calculations
- Power BI dashboard development
- Customer analysis
- Product analysis
- Sales analysis
- Seller analysis
- Delivery analysis
- Business insight generation
- Data-driven recommendations



The project therefore represents a complete **data analyst portfolio case study**, rather than simply a collection of charts.